



Database Management Systems

Week 7, Day 4

MongoDB—Geospatial Databases

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Download Sample Database

- Download the example datasets from
- **Neighbourhoods:**
 - <https://raw.githubusercontent.com/mongodb/docs-assets/geospatial/neighborhoods.json>
- **Restaurants:**
 - <https://raw.githubusercontent.com/mongodb/docs-assets/geospatial/restaurants.json>

Import Collections

MongoDB > restInfoDB > GeoRest

Documents 25.4K

Aggregations

Schema

Indexes 2

Validation



Type a query: { field: 'value' } or [Generate query](#) ✨

➕ ADD DATA ▼

📄 EXPORT DATA ▼

✎ UPDATE

🗑 DELETE

Import JSON or CSV file

Insert document

476c522cafdb053add')

ke Shop"

```
> db.GeoRest.createIndex({location: "2dsphere"})  
< location_2dsphere  
> db.neighborhoods.createIndex({ geometry: "2dsphere" })  
< geometry_2dsphere  
restInfoDB >
```

Create Indexes

- A geospatial index, and almost always improves performance of \$geoWithin and \$geoIntersects queries.

Exploring the Data

Inspect an entry in the newly-created GeoRest collection in mongosh:

```
> db.GeoRest.find()
< {
  _id: ObjectId('55cba2476c522cafdb053add'),
  location: {
    coordinates: [
      -73.856077,
      40.848447
    ],
    type: 'Point'
  },
  name: 'Morris Park Bake Shop'
}
```

Current Neighborhood

- Suppose the user is located at -73.93414657 longitude and 40.82302903 latitude. To find the current neighborhood, you will specify a point using the special \$geometry field in GeoJSON format

```
> db.neighborhoods.findOne(  
  {  
    geometry: {  
      $geoIntersects: {  
        $geometry: {  
          type: "Point",  
          coordinates: [ -73.93414657, 40.82302903 ]  
        }  
      }  
    }  
  }  
)
```

Output

```
{
  "_id" : ObjectId("55cb9c666c522cafdb053a68"),
  "geometry" : {
    "type" : "Polygon",
    "coordinates" : [
      [
        [
          -73.93383000695911,
          40.81949109558767
        ],
        ...
      ]
    ]
  },
  "name" : "Central Harlem North-Polo Grounds"
}
```

Find all Restaurants in the Neighborhood

You can also query to find all restaurants contained in a given neighborhood
(127)

```
restInfoDB> var neighborhood = db.neighborhoods.findOne(  
  { geometry:  
    { $geoIntersects:  
      { $geometry:  
        { type: "Point", coordinates: [ -73.93414657, 40.82302903 ] } } } } )  
db.GeoRest.find( { location: { $geoWithin: { $geometry: neighborhood.geometry } } } ).count()
```

Find Restaurants within a Distance

```
restInfoDB> var METERS_PER_MILE = 1609.34
             db.GeoRest.find(
               { location:
                 { $nearSphere:
                   { $geometry:
                     { type: "Point", coordinates: [ -73.93414657, 40.82302903 ] },
                     $maxDistance: 5 * METERS_PER_MILE } } })
```

Question(s)
